

Tensile Testing of 6061 Aluminum and 4140 Steel

Stress–strain behavior, elastic modulus, and mechanical properties from uniaxial tension tests

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Course	MECH 237A Strength of Materials Lab, NJIT	Instructor	Prof. Milano · TA: Oscar Poudel
Team	Kacper Czarnecki, Ashlee Hercules, Luis Ramirez, Jackson Robillard	My role	Wrote the introduction and theory section; individual discussion and conclusion
Equipment	Universal testing machine, extensometer, vernier caliper		

PROJECT SUMMARY

This experiment characterized how two common engineering metals, 6061 aluminum and 4140 steel, respond to uniaxial tension. Each specimen was measured, fitted with an extensometer, and loaded to failure in a universal testing machine that recorded load and elongation every second. From the resulting stress–strain curves, the team determined the elastic modulus, 0.2% offset yield strength, ultimate strength, modulus of resilience, toughness, and percent elongation, and checked the axial deformation relation $\delta = PL/AE$. The measured elastic modulus was 69.9 GPa for aluminum (1.3% from the 69 GPa reference) and 227.2 GPa for steel (about 13% above the 192–210 GPa reference range). Both materials necked before fracture, confirming ductile failure.

1. Introduction & Theory

Understanding the mechanical behavior of materials is essential to designing safe, efficient, and precise structures and machines, because every material responds differently under stress. One of the most basic ways to evaluate strength is the tension test, which captures the full stress–strain relationship of a material under axial load.

Most engineering metals are treated as homogeneous, with consistent properties throughout. At low stress, a ductile metal follows **Hooke’s law**: stress and strain are linearly related, and the slope of that line is the material’s **Young’s modulus (E)**. As stress increases, the material reaches its **proportional limit**, where the curve stops being linear. Close to this point are the **elastic limit**, the maximum stress the material can take and still return to its original shape, and the **yield point**, where permanent deformation begins.

The stress at yielding is the **yield strength (σ_y)**, a critical design parameter. Many materials, including aluminum alloys and brittle materials, do not show a distinct yield point, so yield strength is defined using the 0.2% offset method: the stress that leaves 0.2% permanent strain after unloading. As stress continues to rise, the material reaches its **ultimate strength**, the maximum stress it can carry. Brittle materials fracture here, while ductile materials keep deforming and neck down until they reach their **rupture strength**.

Stress is calculated from the cross-sectional area ($\sigma = P/A$) and strain from the elongation relative to gauge length ($\epsilon = \delta/L$). Combining these with Hooke’s law ($\sigma = E\epsilon$) gives the axial deformation equation $\delta = PL/AE$. These relationships are fundamental to engineering design because they predict how a material will behave under load while maintaining structural integrity.

2. Procedure

After putting on safety goggles, the team measured and recorded the initial diameter of each specimen. Each specimen was then mounted in the universal testing machine, and an extensometer was attached to record elongation. The specimen was loaded until it necked and fractured with a sudden, loud break, while the machine logged load and deformation. Finally, the final diameter at the fracture was measured and recorded.

3. Results

Table 1. Measured tensile properties (reference: 69 GPa for 6061-T6; 192–210 GPa for 4140)

Property	6061 Aluminum	% error	4140 Steel	% error
Initial diameter (in)	0.506		0.503	
Final diameter (in)	0.3945		0.3265	
Initial area (in ²)	0.2011		0.1987	
Young's modulus	10.13 × 10 ⁶ psi (69.88 GPa)	1.3%	32.94 × 10 ⁶ psi (227.19 GPa)	13.0%
0.2% offset yield strength	46,809 psi (322.7 MPa)	8.9%	139,168 psi (959.5 MPa)	65.7%
Ultimate strength	51,896 psi (357.8 MPa)	5.6%	154,989 psi (1,068.6 MPa)	15.1%
Modulus of resilience	244.1 psi		962.4 psi	
Toughness	9,880 psi		45,634 psi	
Elongation	22.4%		33.0%	
Reduction of area*	39.2%		57.9%	

*Reduction of area was calculated from the measured initial and final diameters: $RA = 1 - (d_f/d_0)^2$.

Elastic modulus: sample calculation

The elastic modulus was also calculated by hand from two well-separated points on the linear portion of each stress–strain curve, using $E = (\sigma_2 - \sigma_1) / (\epsilon_2 - \epsilon_1)$:

Table 2. Two-point elastic modulus calculation

Material	σ_1 (psi)	σ_2 (psi)	ϵ_1	ϵ_2	E (calculated)	Diff. from trendline
6061 Aluminum	10,805.76	38,827.38	0.00104	0.00385	9,972 ksi (68.76 GPa)	–1.6%
4140 Steel	28,049.84	114,218.3	0.00071	0.00333	32,889 ksi (226.76 GPa)	–1.7%

Both hand calculations fall within 1.7% of the trendline slope, which shows the chosen points sit on the linear region and that the two-point method is a sound check on the regression.

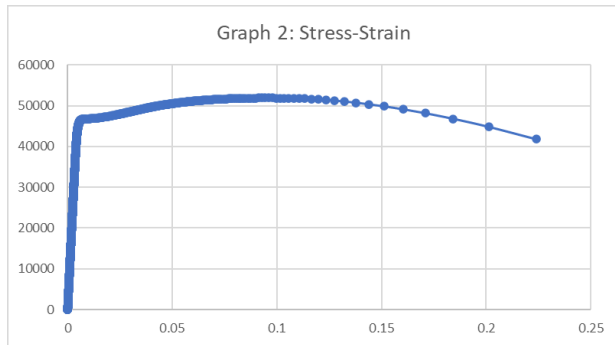


Figure 1. 6061 aluminum stress-strain curve

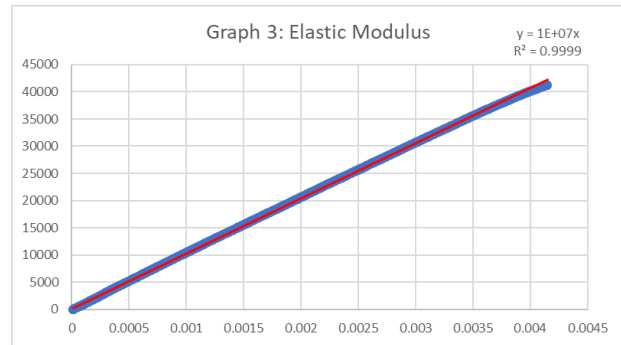


Figure 2. 6061 aluminum elastic region with linear fit

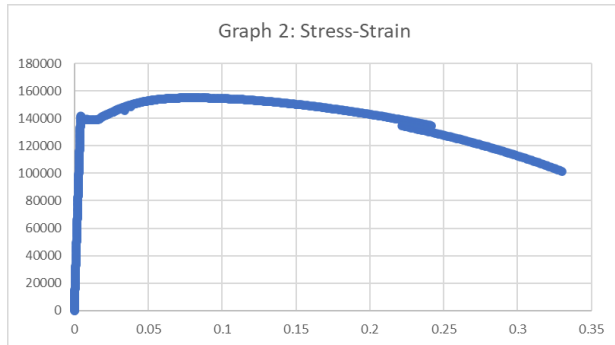


Figure 3. 4140 steel stress-strain curve

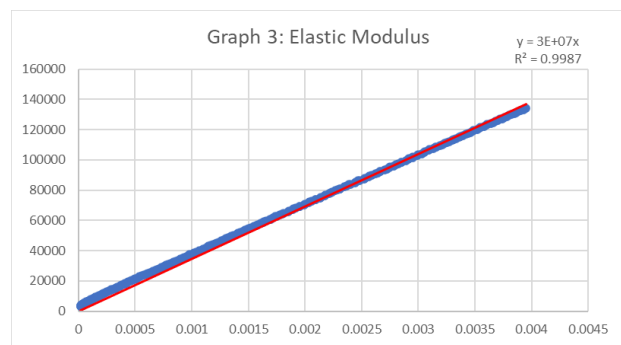


Figure 4. 4140 steel elastic region with linear fit



Figure 5. Steel before test



Figure 6. Steel after failure



Figure 7. Aluminum before test



Figure 8. Aluminum after failure

4. Discussion

Elastic modulus. The main goal of the experiment was to measure the elastic modulus, and the aluminum result was excellent: 69.9 GPa against a 69 GPa reference, only 1.3% off, which suggests the sample and setup were representative of the material. The steel result, 227.2 GPa, was about 13% higher than its reference range. That is close enough to show the expected behavior, but large enough to point to a real issue in the steel run — possibilities include extensometer slip or seating at the start of the test, alignment in the grips, or the specific heat-treated condition of the 4140 bar. Without repeat tests, the exact cause cannot be isolated.

Strength values. The aluminum yield and ultimate strengths were within 9% and 6% of reference values. The steel deviated more, with a 66% error in yield strength and 15% in ultimate strength. 4140 is a heat-treatable alloy whose strength varies widely with its condition, so this bar was likely in a different heat-treated state than the

reference material. Published values are based on large numbers of standardized tests, while this experiment tested a single specimen of each material, so some difference is expected. Testing multiple specimens would give a far more accurate picture of each material's true range.

Ductility and failure. Both specimens necked visibly before fracture, and both stress–strain curves show a long plastic region with strain hardening up to the ultimate stress followed by softening to failure. This confirms ductile behavior; a brittle material would instead fracture suddenly with little plastic deformation. The steel absorbed far more energy before failure, with a toughness of about 45,600 psi versus 9,900 psi for aluminum.

Hooke's law. In the elastic region of both materials, load and deformation were clearly linear, and the load–deformation and stress–strain curves had the same shape. This confirms Hooke's law and the axial deformation equation $\delta = PL/AE$, which describes exactly this linear elastic behavior.

5. Conclusion

The experiment achieved its objectives: it measured the proportional limit, yield strength, ultimate strength, elastic modulus, resilience, and toughness of 6061 aluminum and 4140 steel, and confirmed the linear elastic behavior predicted by Hooke's law. The aluminum modulus matched its reference within 1.3%, while the steel modulus came in about 13% high, most likely due to setup or material-condition effects that a single test cannot resolve. Both materials failed in a ductile manner with clear necking. Overall, the results reinforce why material testing matters in design: engineers can only select materials confidently when they understand how those materials deform and fail under load.

References

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